Name – Bhakare Purva Shivaji

Roll no- 08

Batch – T1

```
# ============================================================
# HEART DISEASE DATASET - DATA PREPROCESSING AND ANALYSIS
# ============================================================
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
# ============================================================
# 1. LOAD HEART DISEASE DATASET
# ============================================================
# IMPORTANT:
# Replace this with the actual CSV file name.
# Example:
# C:\Users\PVGIT\Downloads\archive (6)\heart.csv
file_path = r"C:\Users\PVGIT\Downloads\archive (6)\heart.csv"
df = pd.read_csv(file_path)
print("First 5 rows:")
print(df.head())
# ============================================================
# 2. DATASET INFORMATION
# ============================================================
print("\nDataset Information:")
df.info()
# ============================================================
# 3. HANDLE MISSING VALUES
# ============================================================
# Fill missing numerical values with the mean
```

```
columns_to_fill = [
    "age",
    "trestbps",
    "chol",
    "thalach",
    "oldpeak"
]
for column in columns_to_fill:
    df[column] = df[column].fillna(df[column].mean())
print("\nMissing values after cleaning:")
print(df.isnull().sum())
# ============================================================
# 4. ENCODING
# ============================================================
# The dataset already contains numerical values.
# Therefore, no additional encoding is required.
print("\nEncoding:")
print("All required categorical values are already numerically encoded.")
# ============================================================
# 5. STATISTICAL ANALYSIS
# ============================================================
print("\nStatistical Analysis:")
print(df.describe())
# ============================================================
# 6. SELECT INPUT AND OUTPUT FEATURES
# ============================================================
X = df[
    [
        "age",
        "sex",
        "cp",
```

```
        "trestbps",
        "chol",
        "fbs",
        "restecg",
        "thalach",
        "exang",
        "oldpeak",
        "slope",
        "ca",
        "thal"
    ]
]
y = df["target"]
# ============================================================
# 7. SPLIT DATASET INTO TRAINING AND TESTING DATA
# ============================================================
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
print("\nTraining data shape:", X_train.shape)
print("Testing data shape :", X_test.shape)
print("\nTraining target shape:", y_train.shape)
print("Testing target shape :", y_test.shape)
# ============================================================
# 8. FEATURE SCALING
# ============================================================
scaler = StandardScaler()
# Fit scaler only on training data
```

```
X_train_scaled = scaler.fit_transform(X_train)
# Use the same scaler on test data
X_test_scaled = scaler.transform(X_test)
print("\nScaled Training Data:")
print(X_train_scaled[:5])
print("\nScaled Testing Data:")
print(X_test_scaled[:5])
# ============================================================
# 9. HISTOGRAM
# ============================================================
df.hist(
    figsize=(12, 10),
    bins=20
)
plt.suptitle(
    "Histograms of Heart Disease Dataset",
    fontsize=16
)
plt.tight_layout()
plt.show()
# ============================================================
# 10. BOXPLOT
# ============================================================
plt.figure(figsize=(10, 6))
df[
    [
        "age",
        "trestbps",
        "chol",
        "thalach",
        "oldpeak"
```

```
    ]
].boxplot()
plt.title("Boxplot of Numerical Features")
plt.ylabel("Values")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
# ============================================================
# 11. FINAL INFORMATION
# ============================================================
print("\n========================================")
print("DATA PREPROCESSING COMPLETED")
print("========================================")
print("Original dataset shape :", df.shape)
print("Training data shape    :", X_train_scaled.shape)
print("Testing data shape     :", X_test_scaled.shape)
print("Number of input features:", X.shape[1])
print("\nProgram executed successfully!")
```

OUTPUT-

UTPUT-First 5 rows:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	0
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
3	61	1	0	148	203	0	1	161	0	0.0	2			
4	62	0	0	138	294	1	1	106	0	1.9	1			

3 1 3 0

4 3 2 0

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 1025 entries, 0 to 1024

Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	age	1025 non-null	int64
1	sex	1025 non-null	int64
2	cp	1025 non-null	int64
3	trestbps	1025 non-null	int64
4	chol	1025 non-null	int64
5	fbs	1025 non-null	int64
6	restecg	1025 non-null	int64
7	thalach	1025 non-null	int64
8	exang	1025 non-null	int64
9	oldpeak	1025 non-null	float64
10	slope	1025 non-null	int64
11	ca	1025 non-null	int64
12	thal	1025 non-null	int64
13	target	1025 non-null	int64

dtypes: float64(1), int64(13)

memory usage: 112.2 KB

Missing values after cleaning:

age 0

sex 0

cp 0

trestbps 0

chol 0

fbs 0

restecg 0

thalach 0

exang 0

oldpeak 0

slope 0

ca 0

thal 0

target 0

dtype: int64

Encoding:

All required categorical values are already numerically encoded.

Statistical Analysis:

	age	sex	cp	trestbps	chol \
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.00000
mean	54.434146	0.695610	0.942439	131.611707	246.00000
std	9.072290	0.460373	1.029641	17.516718	51.59251
min	29.000000	0.000000	0.000000	94.000000	126.00000
25%	48.000000	0.000000	0.000000	120.000000	211.00000
50%	56.000000	1.000000	1.000000	130.000000	240.00000
75%	61.000000	1.000000	2.000000	140.000000	275.00000
max	77.000000	1.000000	3.000000	200.000000	564.00000

	fbs	restecg	thalach	exang	oldpeak \
count	1025.000000	1025.000000	1025.000000	1025.000000	1025.000000
mean	0.149268	0.529756	149.114146	0.336585	1.071512
std	0.356527	0.527878	23.005724	0.472772	1.175053
min	0.000000	0.000000	71.000000	0.000000	0.000000
25%	0.000000	0.000000	132.000000	0.000000	0.000000
50%	0.000000	1.000000	152.000000	0.000000	0.800000

```
75%      0.000000    1.000000   166.000000    1.000000    1.800000
max      1.000000    2.000000   202.000000    1.000000    6.200000

           slope          ca        thal      target
count  1025.000000  1025.000000  1025.000000  1025.000000
mean      1.385366     0.754146     2.323902     0.513171
std       0.617755     1.030798     0.620660     0.500070
min       0.000000     0.000000     0.000000     0.000000
25%       1.000000     0.000000     2.000000     0.000000
50%       1.000000     0.000000     2.000000     1.000000
75%       2.000000     1.000000     3.000000     1.000000
max       2.000000     4.000000     3.000000     1.000000

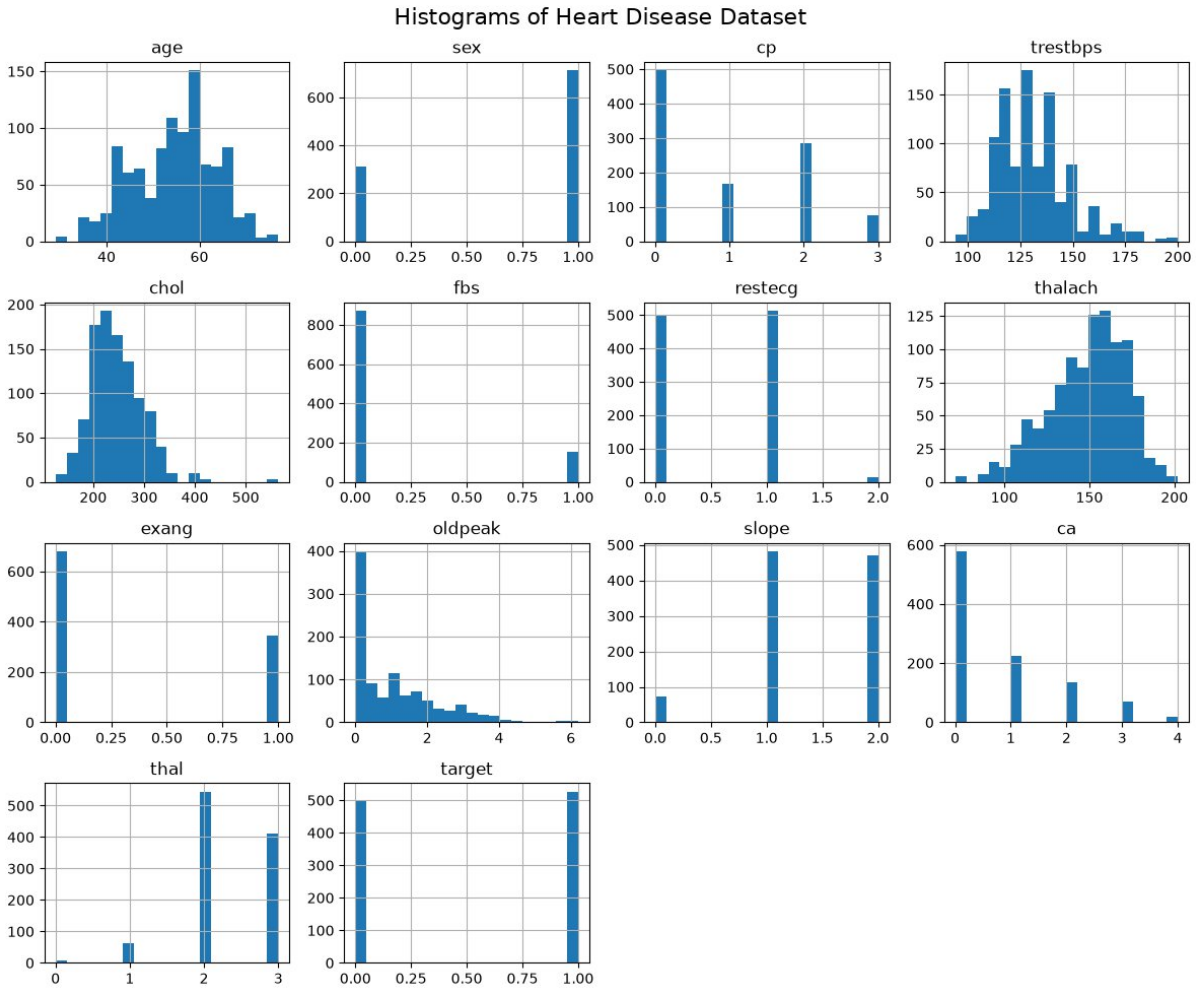
Training data shape: (820, 13)
Testing data shape : (205, 13)
Training target shape: (820,)
Testing target shape : (205,)

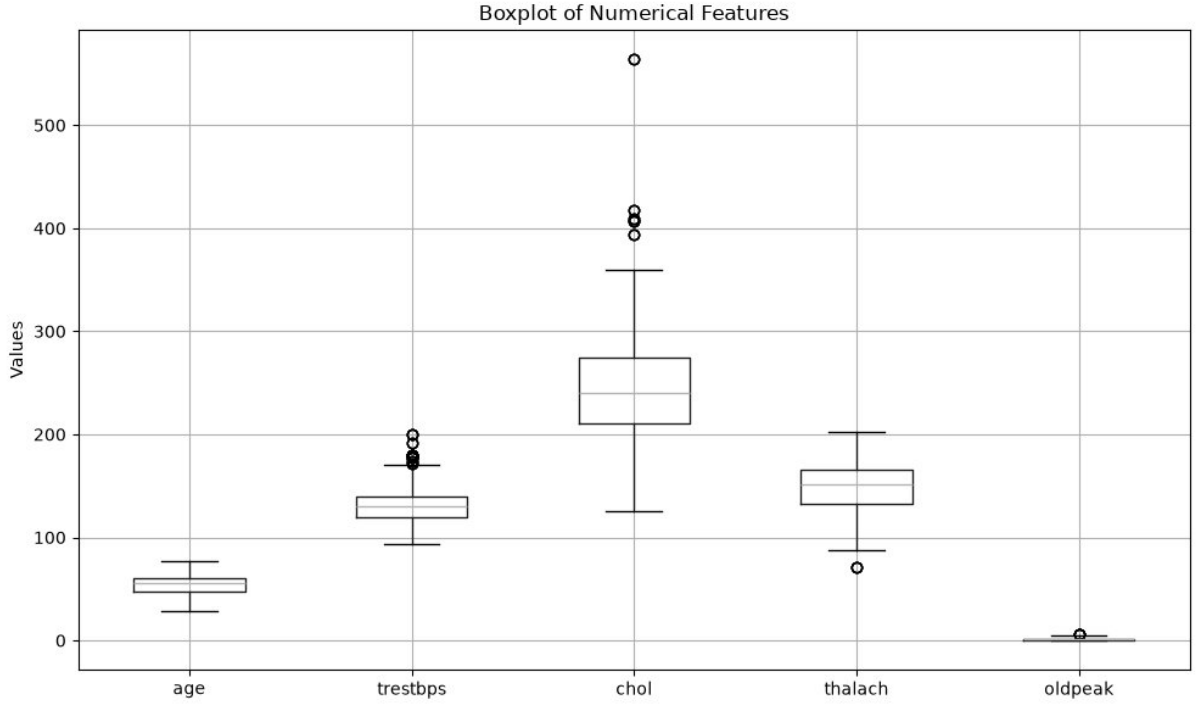
Scaled Training Data:
[[-5.85840222e-01  6.54653671e-01  1.00827500e+00 -7.79453566e-01
  -1.93503098e+00 -4.14039336e-01 -9.83741901e-01 -1.01909426e+00
  -7.25948939e-01 -2.10661213e-01  1.00526437e+00  2.17169136e+00
  -5.45193165e-01]
 [ 1.05147737e+00 -1.52752523e+00 -9.16720340e-01  2.74173173e+00
    1.61063407e+00 -4.14039336e-01  9.09845796e-01  2.02882145e-01
    1.37750735e+00 -9.12152360e-01  1.00526437e+00 -7.25467395e-01
  -5.45193165e-01]
 [-4.00676907e-02 -1.52752523e+00  1.00827500e+00 -1.34738668e+00
    4.42176271e-01 -4.14039336e-01 -9.83741901e-01  7.70228333e-01
  -7.25948939e-01 -9.12152360e-01  1.00526437e+00 -7.25467395e-01
  -5.45193165e-01]
```

[5.05704840e-01 6.54653671e-01 -9.16720340e-01 1.86032724e-01
 -2.22635925e-01 -4.14039336e-01 9.09845796e-01 5.08376246e-01
 -7.25948939e-01 -4.73720393e-01 -6.40078509e-01 -7.25467395e-01
  1.11057867e+00]
[-3.67531209e-01 6.54653671e-01 1.00827500e+00 -3.81900388e-01
 -1.03185928e-03 2.41522946e+00 -9.83741901e-01 7.26586318e-01
 -7.25948939e-01 1.19232108e+00 -6.40078509e-01 -7.25467395e-01
 -5.45193165e-01]]

Scaled Testing Data:

[[0.83316836 -1.52752523 -0.91672034 -0.4386937 -0.72628153 -0.41403934
  0.9098458   0.59566027 -0.72594894 -0.91215236 1.00526437 -0.7254674
 -0.54519316]
[-0.1492222 -1.52752523 1.008275   -0.21152045 -0.58526076 -0.41403934
 -0.9837419 -1.49915642 -0.72594894 -0.91215236 1.00526437 -0.7254674
 -3.85673683]
[0.06908682 0.65465367 -0.91672034 1.6058655   0.8853844 -0.41403934
 -0.9837419 -0.18989598 1.37750735 -0.21066121 -0.64007851 0.24025219
  1.11057867]
[-0.47668572 -1.52752523 0.04577733 -0.66586694 -0.02117768 -0.41403934
  0.9098458   0.55201826 -0.72594894 0.05239797 1.00526437 -0.7254674
 -0.54519316]
[-0.69499473 0.65465367 -0.91672034 -0.09793383 0.22057221 2.41522946
 -0.9837419   0.02831409 1.37750735 -0.91215236 1.00526437 1.20597177
  1.11057867]]


Histograms of Heart Disease Dataset
age
sex
cp
trestbps
chol
fbs
restecg
thalach
exang
oldpeak
slope
ca
thal
target


Boxplot of Numerical Features
Values
age
trestbps
chol
thalach
oldpeak

==========

DATA PREPROCESSING COMPLETED

==========

Original dataset shape : (1025, 14)

Training data shape : (820, 13)

Testing data shape : (205, 13)

Number of input features: 13

Program executed successfully!